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GRAHAM DIB

MATH 70095 - Applicable Maths

Autumn 2025 - Assessment 3

Deadline: 11 December 2025, 09:00 (UK time)

You should submit a PDF document containing your answers to these questions, via the Blackboard VLE, by the deadline stated above. Your submission may be a scanned copy of handwritten answers, or a typed document; all submissions should be clear and legible, and should also ***show all of your working***. This coursework should involve approximately **6 hours** of effort. The available marks are indicated in square brackets for each question.

This coursework counts for 50% of your total mark for Applicable Maths.

This assignment must be attempted individually; your submission must be your own, unaided work. Candidates are prohibited from discussing assessed coursework, and must abide by [Imperial College's rules](#) regarding academic integrity and plagiarism. Unless specifically authorised within the assignment instructions, the submission of output from [generative AI tools](#) (e.g., ChatGPT) for assessed coursework is prohibited. Violations will be treated as an examination offence. Enabling other candidates to plagiarise your work constitutes an examination offence. To ensure quality assurance is maintained, departments may choose to invite a random selection of students to an 'authenticity interview' on their submitted assessments.

Q1) *There are up to two marks available in this question for clarity of exposition.*

A random variable Φ is said to have a *Cardioid* distribution with frequency $k \in \mathbb{N}$ (where here $\mathbb{N} := \{1, 2, 3, \dots\}$ is the set of positive integers) and shape parameter $\rho \in \mathbb{R}$ satisfying $|\rho| < \frac{1}{2}$, if its probability density function has the form

$$f_{\Phi}(\phi) = \frac{1}{2\pi}(1 + 2\rho \cos(k\phi)), \quad \phi \in [0, 2\pi).$$

In this question, you may use without proof that for such a random variable, for all $\ell \in \mathbb{N}$,

$$\mathbb{E}[\cos(\ell\Phi)] = \begin{cases} \rho, & \ell = k \\ 0, & \ell \neq k \end{cases} \quad \text{and} \quad \mathbb{E}[\sin(\ell\Phi)] = 0.$$

Let $\{X_t\}_{t \in \mathbb{R}}$ be the stochastic process defined by

$$X_t := R \cos(\omega t + \Phi), \quad t \in \mathbb{R}$$

where

- $\omega > 0$ is fixed;
- $R \geq 0$ is a non-negative random variable with mean $\mu := \mathbb{E}[R]$ and variance $\sigma^2 := \text{Var}[R]$ satisfying $0 < \mu, \sigma^2 < \infty$;

- Φ is a Cardioid random variable with frequency $k \in \mathbb{N}$ and shape ρ with $|\rho| < \frac{1}{2}$, which is independent of R .

- For all values of k, ρ , find the mean $\mathbb{E}[X_t]$ of $\{X_t\}_{t \in \mathbb{R}}$. [4]
- For all values of k, ρ , find the autocovariance $\text{Cov}[X_t, X_s]$ of $\{X_t\}_{t \in \mathbb{R}}$. [6]
- For what values of k, ρ (if any) is the process weakly stationary? [2]
- For what values of k, ρ (if any) is the process strongly stationary? [4]

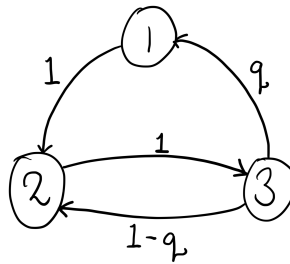
Hint: for this question, you may find the following trigonometric identities useful: for all $a, b \in \mathbb{R}$,

$$(1) \cos(a + b) = \cos(a) \cos(b) - \sin(a) \sin(b)$$

$$(2) \cos(a) \cos(b) = \frac{1}{2} [\cos(a - b) + \cos(a + b)]$$

[18]

Q2) Consider the time-homogeneous Markov chain $\{X_t\}_{t \in \mathbb{N}}$ with the following transition diagram:



Correction: (29 November 2025) The transition probability from state 3 to state 2 previously read q where it should have been $1 - q$. This has been corrected in this version.

- For what values of $q \in (0, 1]$ is $\{X_t\}_{t \in \mathbb{N}}$ irreducible. For what values of $q \in [0, 1]$ does it have a unique stationary distribution? Justify your answer. [2]
- For all values of $q \in (0, 1]$ for which $\{X_t\}_{t \in \mathbb{N}}$ has a unique stationary, compute it. [2]
- For what values of $q \in (0, 1]$ will the sequence of marginal distributions $\pi^{(t)}$ of $\{X_t\}_{t \in \mathbb{N}}$ converge to a unique stationary distribution for all initial distributions $\pi^{(0)}$? [3]

[7]

Q3) There are up to two marks available in this question for clarity of exposition.

Let $X_1, \dots, X_n$ be independent and identically distributed random variables from the $\text{Uniform}(0, \theta)$ distribution, where $\theta > 0$ is an unknown parameter.

- Prove that $M := \max_{i \in \{1, \dots, n\}} X_i$ is a sufficient statistic for θ . Is $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ a sufficient statistic for θ ? Briefly justify your answer. [3]

- (b) Write down the likelihood $L(\theta | x)$ of θ and use it to derive the maximum likelihood estimator $\hat{\theta}_{\text{MLE}}$ of θ . [3]
- (c) Derive the method-of-moments estimator $\hat{\theta}_{\text{MoM}}$ of θ . [2]
- (d) Derive the CDF of $\hat{\theta}_{\text{MLE}}$ and use it to construct an interval estimator for θ from $\hat{\theta}_{\text{MLE}}$ with exactly 95% coverage probability. [4]
- (e) Use the central limit theorem to derive an interval estimator for θ from $\hat{\theta}_{\text{MoM}}$ with approximately 95% coverage probability. [4]

You may use without proof that for a random variable $X \sim \text{Uniform}(0, \theta)$,

$$\mu := \mathbb{E}[X] = \frac{\theta}{2}, \quad \sigma^2 := \text{Var}[X] = \frac{\theta^2}{12}.$$

For a random variable $Z \sim N(0, 1)$, the table below shows the values of the quantiles z_α such that

$$P(Z \leq z_\alpha) = \alpha,$$

for selected values of α .

α	0.9	0.95	0.975	0.99	0.995	0.999
z_α	1.282	1.645	1.960	2.326	2.576	3.090

[18]

- Q4) Three widely used measures of distance between probability distributions are the *Total Variation*, the *Kullback-Leibler divergence* and the *Hellinger distance*. Consider two probability measures P, Q defined on a common measurable space $(\Omega, \mathcal{F})$. The *Total-Variation* is defined as

$$\text{TV}(P, Q) := \sup_{A \in \mathcal{F}} |P(A) - Q(A)|.$$

If Ω is discrete and $P(x), Q(x) > 0$ for all $x \in \Omega$, then the *Kullback-Leibler divergence* is defined as

$$\text{KL}(P, Q) = \sum_{x \in \Omega} P(x) \log \frac{P(x)}{Q(x)}.$$

If P, Q are absolutely continuous with respect to some reference measure, so that they have probability density function p, q , respectively, then the *squared Hellinger distance* is defined as

$$H^2(P, Q) = \frac{1}{2} \int_{\Omega} (\sqrt{p(x)} - \sqrt{q(x)})^2 dx.$$

Answer the following questions assuming the settings in which each of the measures of distance are defined above.

- a) Is Total Variation a metric? Justify your answer. [3]
- b) Is Kullback-Leibler a metric? Justify your answer. [2]
- c) By giving a counter example, show that the squared Hellinger distance is *not* a metric. [3]

[8]

Q1) a)

$$X_t = R \cos(\omega t + \Phi), \quad t \in \mathbb{R}, \quad R \text{ independent of } \Phi,$$

$$E[R] = \mu \quad \text{and} \quad \Phi \sim \text{circular}(\kappa, \rho)$$

$$X_t = R \cos(\omega t + \Phi), \text{ where } R \text{ and } \Phi \text{ are random variables,}$$

$\therefore \{X_t\}_{t \in \mathbb{R}}$ is a continuous stochastic process.

Computing $E[X_t]$:

$$E[X_t] = E[R \cos(\omega t + \Phi)] = E[R] \cdot E[\cos(\omega t + \Phi)] \quad (\text{as } R \text{ is independent of } \Phi)$$

$$E[R] \cdot E[\cos(\omega t + \Phi)] = \mu \cdot E[\cos(\omega t + \Phi)] \quad (\text{as } E[R] = \mu)$$

Recall the trigonometric addition identity for cosine:

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta \quad \text{for all } \alpha, \beta \in \mathbb{R} \quad \textcircled{1}$$

consider $E[\cos(\omega t + \Phi)]$, and apply identity $\textcircled{1}$ to the expression:

$$E[\cos(\omega t + \Phi)] = E[\cos \omega t \cos \Phi - \sin \omega t \sin \Phi]$$

$\omega > 0$ is fixed and $t \in \mathbb{R}$, so $E[\cos \omega t] = \cos \omega t$ and $E[\sin \omega t] = \sin \omega t$ (ω and t are not random variables)

$$\therefore E[\cos(\omega t + \Phi)] = E[\cos \omega t \cos \Phi - \sin \omega t \sin \Phi] = \cos \omega t E[\cos \Phi] - \sin \omega t E[\sin \Phi]$$

$$\therefore E[X_t] = \mu \cdot E[\cos(\omega t + \Phi)] = \mu \cdot [\cos \omega t E[\cos \Phi] - \sin \omega t E[\sin \Phi]]$$

From Q1, we can use without proof that for such a random variable, for all $\ell \in \mathbb{N}$,

$$E[\cos(\ell \Phi)] = \begin{cases} \rho, & \ell = \kappa \\ 0, & \ell \neq \kappa \end{cases} \quad \text{and} \quad E[\sin \ell \Phi] = 0 \quad \textcircled{2}$$

$$\therefore E[X_t] = \mu \cdot \cos \omega t \cdot E[\cos \ell \Phi], \text{ where } \ell = 1 \quad (\text{using } \textcircled{2}, E[\sin \ell \Phi] = 0)$$

$$\therefore E[X_t] = \begin{cases} \mu \cdot \rho \cdot \cos \omega t, & \text{if } k=1 \\ 0, & \text{if } k \neq 1 \end{cases} \quad \omega > 0, t \in \mathbb{R} \quad (\text{using } ②)$$

b) Recall the definition of autocovariance:

$$\text{cov}[X_t, X_s] = E[X_t X_s] - E[X_t]E[X_s]$$

First consider and compute $E[X_t X_s]$

$$\begin{aligned} E[X_t X_s] &= E[(R \cos(\omega t + \Phi)) \cdot (R \cos(\omega s + \Phi))] = E[R^2 \cdot \cos(\omega t + \Phi) \cos(\omega s + \Phi)] \\ &= E[R^2] \cdot E[\cos(\omega t + \Phi) \cos(\omega s + \Phi)] \quad (\text{by independence of } R) \end{aligned}$$

$$\text{Recall } E[R] = \mu \quad \text{and} \quad \text{Var}[R] = E[R^2] - E[R]^2 = \sigma^2$$

$$E[R^2] - E[R]^2 = E[R^2] - \mu^2 = \sigma^2$$

$$\therefore E[R^2] = \mu^2 + \sigma^2$$

$$E[X_t X_s] = (\mu^2 + \sigma^2) \cdot E[\cos(\omega t + \Phi) \cos(\omega s + \Phi)]$$

Using the trigonometric product identity of cosine,

$$\cos(a) \cos(b) = \frac{1}{2} [\cos(a-b) + \cos(a+b)] \quad \text{for all } a, b \in \mathbb{R}$$

$$\begin{aligned} E[X_t X_s] &= (\mu^2 + \sigma^2) \cdot E\left[\frac{1}{2} [\cos(\omega t + \Phi - \omega s - \Phi) + \cos(\omega t + \Phi + \omega s + \Phi)]\right] \\ &= (\mu^2 + \sigma^2) \cdot E\left[\frac{1}{2} [\cos(\omega t - \omega s) + \cos(\omega t + \omega s + 2\Phi)]\right] \\ &= (\mu^2 + \sigma^2) \cdot \frac{1}{2} [\cos(\omega t - \omega s) + E[\cos(\omega t + \omega s + 2\Phi)]] \end{aligned}$$

Consider and expand the last term using the trigonometric addition identity of cosine,

$$\cos(a+b) = \cos(a)\cos(b) - \sin(a)\sin(b) \quad \text{for all } a, b \in \mathbb{R}$$

$$\cos(\omega(t+s) + 2\Phi) = \cos(\omega(t+s))\cos(2\Phi) - \sin(\omega(t+s))\sin(2\Phi)$$

$$\begin{aligned} E[X_t X_s] &= (\mu^2 + \sigma^2) \cdot \frac{1}{2} [\cos(\omega t - \omega s) + E[\cos(\omega t + \omega s + 2\Phi)]] \\ &= (\mu^2 + \sigma^2) \cdot \frac{1}{2} [\cos(\omega(t-s)) + E[\cos(\omega(t+s))\cos(2\Phi) - \sin(\omega(t+s))\sin(2\Phi)]] \\ &= (\mu^2 + \sigma^2) \cdot \frac{1}{2} [\cos(\omega(t-s)) + \cos(\omega(t+s))E[\cos(2\Phi)] - \sin(\omega(t+s))E[\sin(2\Phi)]] \end{aligned}$$

Recall that $E[\sin(\lambda \Phi)] = 0$ for all λ , therefore $E[\sin(2\Phi)] = 0$

$$E[X_t X_s] = (u^2 + \sigma^2) \cdot \frac{1}{2} [\cos(\omega(t-s)) + \cos(\omega(t+s)) E(\cos(2\Phi))]]$$

$$\text{Recall that } E(\cos(\lambda \Phi)) = \begin{cases} \rho, & \lambda = k \\ 0, & \lambda \neq k \end{cases}$$

$$\therefore E[X_t X_s] = (u^2 + \sigma^2) \cdot \frac{1}{2} [\cos(\omega(t-s)) + \cos(\omega(t+s)) \cdot \rho] \quad \text{where } \lambda = 2$$

and therefore

$$E[X_t X_s] = \begin{cases} \frac{\sigma^2 + u^2}{2} [\cos(\omega(t-s)) + \rho \cos(\omega(t+s))] , & \text{if } k=2 \\ 0 , & \text{if } k \neq 2 \end{cases} \quad \omega > 0, t \in \mathbb{R}$$

Recall the definition of autocovariance:

$$\text{cov}[X_t, X_s] = E[X_t X_s] - E[X_t] E[X_s]$$

from part (a)

$$E[X_t] = \begin{cases} \mu \rho \cos(\omega t), & k=1 \\ 0, & k \neq 1 \end{cases}$$

using this:

$$E[X_t] E[X_s] = \mathbb{1}_{k=1} \cdot \mu \rho \cos(\omega t) \cdot \mu \rho \cos(\omega s) = \mathbb{1}_{k=1} \mu^2 \rho^2 \cos(\omega t) \cos(\omega s) \quad \text{where } \mathbb{1}_{k=n} \text{ is the indicator function.}$$

$$\therefore \text{cov}[X_t, X_s] = E[X_t X_s] - E[X_t] E[X_s] =$$

$$\frac{\sigma^2 + u^2}{2} [\cos(\omega(t-s)) + \mathbb{1}_{k=2} \rho \cos(\omega(t+s))] - \mathbb{1}_{k=1} \mu^2 \rho^2 \cos(\omega t) \cos(\omega s)$$

c) Recall definition of weakly stationary from the notes:

Definition 7.1.5. Let $\{X_t\}_{t \in T}$ be a stochastic process. $\{X_t\}_{t \in T}$ is defined to be *second-order* or *weakly stationary* if it has finite first and second moments, and its expectation and autocovariance function are both constant with respect to $t \in T$:

$$\begin{aligned}\mathbb{E}[X_{t_1}] &= \mathbb{E}[X_{t_2}] \quad \forall t_1, t_2 \in T \\ C_{XX}(t, t+h) &= C_{XX}(0, h) = c(h) \quad \forall t \in T, h > 0,\end{aligned}$$

for some real-valued function $c: [0, \infty) \mapsto \mathbb{R}$.

First we must check if the first & second moments of X_t are finite.

$$X_t = R \cos(\omega t + \Phi)$$

First moment:

$$|X_t| = |R \cos(\omega t + \Phi)| \leq R \quad \text{a) } \cos(\omega t + \Phi) \text{ is bounded by 1 and -1.}$$

so $E[|X_t|] \leq E[R] \leq \mu < \infty$ as $E[R] = \mu < \infty$, so the first moment $E[X_t]$ is finite as it is bounded

Second moment:

$$X_t^2 = R^2 \cos^2(\omega t + \Phi) \leq R^2 \text{ as } \cos^2(\omega t + \Phi) \text{ is bounded by 1 and 0}$$

$$E[X_t^2] \leq E[R^2] \leq \sigma^2 + \mu^2 < \infty \text{ as } E[R^2] = \sigma^2 + \mu^2 < \infty, \text{ so the second moment } E[X_t^2] \text{ is finite as it is bounded}$$

Now check when expectation is constant with respect to $t \in \mathbb{R}$

Recall from 1a)

$$E[X_t] = \begin{cases} \mu \cdot p \cdot \cos \omega t, & \text{if } k=1 \\ 0, & \text{if } k \neq 1 \end{cases} \quad \omega > 0, t \in \mathbb{R}$$

so expectation is constant in t iff $k \neq 1$ or $p=0$

Now check when autocovariance function is constant with respect to t

from part 1c):

$$\text{cov}[X_t, X_s] = E[X_t X_s] - E[X_t]E[X_s] =$$

$$\frac{\sigma^2 + \mu^2}{2} \left[\cos(\omega(t-s)) + \frac{1}{k-2} p \cos(\omega(t+s)) \right] - \frac{1}{k-1} \mu^2 p^2 \cos(\omega t) \cos(\omega s)$$

substitute X_s for X_{t+h}

$$\text{cov}[X_t, X_{t+h}] = \frac{\sigma^2 + \mu^2}{2} \left[\cos(-\omega h) + \frac{1}{k-2} p \cos(\omega(2t+h)) \right] - \frac{1}{k-1} \mu^2 p^2 \cos(\omega t) \cos(\omega(t+h))$$

$$= \frac{\sigma^2 + \mu^2}{2} \left[\cos(\omega h) + \frac{1}{k-2} p \cos(\omega(2t+h)) \right] - \frac{1}{k-1} \mu^2 p^2 \cos(\omega t) \cos(\omega(t+h)) \quad (\text{as } \cos(x) = \cos(-x))$$

Recall that expectation is constant in t iff $k \neq 1$ or $p = 0$

now check conditions for expectation cases.

case $k \neq 1$:

if $k \neq 1$, then

$$\text{cov}[X_t, X_{t+h}] = \frac{\sigma^2 + \mu^2}{2} \left[\cos(\omega h) + \frac{1}{k-2} p \cos(\omega(2t+h)) \right]$$

$k \in \mathbb{N}$, so we do not check $k=0$ (as it is not in the domain of k)

if $k=2$, and $p \neq 0$, then the process is not weakly stationary, as term $p \cos(\omega(2t+h))$ depends on t

if $k=2$, and $p=0$ then the process is weakly stationary as $\text{cov}(X_t, X_{t+h}) = \frac{\sigma^2 + \mu^2}{2} (\cos(\omega h))$, which is independent of t

if $k \neq 2$, then the process is weakly stationary as $\text{cov}(X_t, X_{t+h}) = \frac{\sigma^2 + \mu^2}{2} (\cos(\omega h))$, which is independent of t

so for $k \neq 1$, the process is weakly stationary if $k \geq 3$ or $k=2$ and $p=0$

case $k=1$ and $p=0$ (expectation condition)

$$\text{cov}(X_t, X_{t+h}) = \frac{\sigma^2 + \mu^2}{2} \cos(\omega h) \text{ therefore the process is weakly stationary as}$$

combining the conditions: autocovariance is independent of t .

$\therefore$ the process $\{X_t\}_{t \in \mathbb{R}}$ is weakly stationary when $p=0$ or for all $k \geq 3$ when $p \neq 0$, for all $k \in \mathbb{N}$

Q1)d) Recall the definition of strong stationarity (Def 7.1.6 of the lecture notes)

Definition 7.1.6. Let $\{X_t\}_{t \in T}$ be a stochastic process. $\{X_t\}_{t \in T}$ is defined to be *strongly or strictly stationary* if its finite-dimensional distributions $P_{t_1, \dots, t_n}^X$ are invariant under constant shifts in their indices. That is, for any finite index subset $(t_1, \dots, t_n) \subseteq T$, and for all $h \in T$ such that $(t_1 + h, \dots, t_n + h) \subseteq T$,

$$P_{t_1, \dots, t_n}^X = P_{t_1+h, \dots, t_n+h}^X.$$

Our process is:

$$X_t = R \cos(\omega t + \Phi)$$

Evaluate the shifted process:

For any shift $h \in \mathbb{R}$,

$$X_{t+h} = R \cos(\omega(t+h) + \Phi) = R \cos(\omega t + (\omega h + \Phi))$$

A shift in time does not affect R , the amplitude of the wave, so we must evaluate for rotation of the wave.

This is a shift in the cosine wave, so we can write it in the same form and evaluate where the distributions are invariant, i.e.:

$$P_{t_1, \dots, t_n}^X = P_{t_1+h, \dots, t_n+h}^X$$

let $\Phi_n = (\omega h + \Phi) \bmod 2\pi$, the shifted phase.

$$\text{for } P_{t_1, \dots, t_n}^X = P_{t_1+h, \dots, t_n+h}^X$$

$$\text{we require } X_{t_1}, \dots, X_{t_n} \stackrel{d}{=} X_{t_1+h}, \dots, X_{t_n+h}$$

$$= R \cos(\omega t_1 + \Phi), \dots, R \cos(\omega t_n + \Phi) \stackrel{d}{=} R \cos(\omega t_1 + \Phi_n), \dots, R \cos(\omega t_n + \Phi_n)$$

$$\therefore \Phi \stackrel{d}{=} \Phi_n \text{ for all } n$$

Recall the probability density function of Φ :

$$f_{\Phi}(\phi) = \frac{1}{2\pi} (1 + 2\rho \cos(k\phi))$$

for $\Phi \stackrel{d}{=} \Phi_n$, we need

$$f_{\Phi}(\phi) = f_{\Phi}(\phi + \omega h) \text{ for all } \phi, h$$

$$\frac{1}{2\pi} (1 + 2\rho \cos(k\phi)) = \frac{1}{2\pi} (1 + 2\rho \cos(k(\phi + \omega h))) \text{ for all } \phi, h$$

this reduces to:

$$\cos(k\phi) = \cos(k(\phi + nh)) \quad \text{for all } \phi, n$$

This is a rotation in the phase of the cosine wave, so this only holds if the cosine wave is rotation-invariant, i.e. the coefficient of the wave is 0.

$$\therefore p = 0$$

When $p = 0$, the density function is

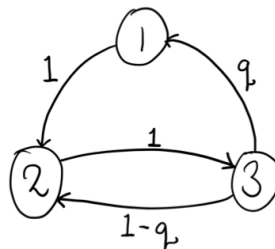
$$f_{\Phi}(\phi) = \frac{1}{2\pi}$$

which is the probability density function of the uniform distribution, or $\Phi \sim \text{Uniform}[0, 2\pi)$

$\therefore \{X_k\}_{k \in \mathbb{N}}$ is strongly stationary for $p = 0$ for all h, k

Q2)

Q2) Consider the time-homogeneous Markov chain $\{X_t\}_{t \in \mathbb{N}}$ with the following transition diagram:



Correction: (29 November 2025) The transition probability from state 3 to state 2 previously read q where it should have been $1 - q$. This has been corrected in this version.

- (a) For what values of $q \in (0, 1]$ is $\{X_t\}_{t \in \mathbb{N}}$ irreducible. For what values of $q \in [0, 1]$ does it have a unique stationary distribution? Justify your answer. [2]
- (b) For all values of $q \in (0, 1]$ for which $\{X_t\}_{t \in \mathbb{N}}$ has a unique stationary, compute it. [2]
- (c) For what values of $q \in (0, 1]$ will the sequence of marginal distributions $\pi^{(t)}$ of $\{X_t\}_{t \in \mathbb{N}}$ converge to a unique stationary distribution for all initial distributions $\pi^{(0)}$? [3]

[7]

a) Recall the definition of irreducible Markov chains (def 7.2.8 from the lecture notes)

Definition 7.2.8. Consider a Markov chain with transition matrix $P = [p_{ij}]$, taking values on the discrete state space S . A subset of the state space, $C \subseteq S$, is said to be *closed* if all states in C communicate only with other states in C , i.e. if $p_{ij} = 0$ for all $i \in C, j \notin C$. We say that C is *irreducible* if all states in C communicate with each other, i.e. $i \longleftrightarrow j$ for all $i, j \in C$.

Define the transitions using the transition matrix.

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ q & 1-q & 0 \end{pmatrix}$$

Find the valid paths (communication)

• 1 and 2

$$1 \rightarrow 2: P_{12} = 1$$

$$2 \rightarrow 1: 2 \rightarrow 3 \rightarrow 1, \text{ probability } P_{23} P_{31} = 1 \cdot q = q$$

$$\therefore 1 \leftrightarrow 2 \text{ if } q > 0$$

• 2 and 3

$$2 \rightarrow 3: P_{23} = 1$$

$$3 \rightarrow 2: \text{if } q < 1, P_{32} = 1-q$$

$$\text{if } q = 1, 3 \rightarrow 1 \rightarrow 2, \text{ probability } P_{31} P_{12} = q \cdot 1 = 1 \cdot 1 = 1$$

$$\therefore 2 \leftrightarrow 3$$

• 1 and 3

$$1 \rightarrow 3: 1 \rightarrow 2 \rightarrow 3, \text{ prob } P_{12} P_{23} = 1 \cdot 1 = 1$$

$$3 \rightarrow 1: P_{31} = q$$

$$\therefore 1 \leftrightarrow 3 \text{ if } q > 0$$

For $\{X_t\}_{t \in \mathbb{N}}$ to be irreducible, the discrete state space $S = \{1, 2, 3\}$ must be irreducible, so all states must communicate with each other, i.e. $i \leftrightarrow j$ for all $i, j \in S$.

We have shown that for communication between all states to be valid, $q > 0$.

$\therefore \{X_t\}_{t \in \mathbb{N}}$ is irreducible for all values of $q \in (0, 1]$

Now check for uniquely stationary:

Recall theorem 7.2.3 from the lecture note:

Theorem 7.2.3. Let $X = \{X_t\}_{t \in \mathbb{N}}$ be a Markov chain defined on the discrete state space S . If S is finite and X is irreducible, then X admits a unique stationary distribution.

since $S = \{1, 2, 3\}$ is finite, and we have shown $\{X_t\}_{t \in \mathbb{N}}$ is irreducible, X admits a uniquely stationary distribution for $q \in (0, 1]$.

Now check for $g=0$:

$$p = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

From state 1: $1 \rightarrow 2$ with prob 1

From state 2: $2 \rightarrow 3$ with prob 1

From state 3: $3 \rightarrow 2$ with prob 1

since $1 \rightarrow 2$, $1 \rightarrow 3$ but $2 \nrightarrow 1$ and $3 \nrightarrow 1$, 1 is a transient state.

since $2 \rightarrow 3$, but $2 \nrightarrow 1$ and $3 \nrightarrow 1$, $\{2, 3\}$ is a closed irreducible class.

Finally, since we only have one closed irreducible class/positive recurrent. This is because the chain will eventually be 'stuck' in the class, and so the long term probabilities will form a unique stationary distribution on this class.

$\therefore$ There exists a uniquely stationary distribution for $\sum_{n=0}^{\infty} p^n$ for all $q \in [0, 1]$

b)

(b) For all values of $q \in (0, 1]$ for which $\{X_t\}_{t \in \mathbb{N}}$ has a unique stationary, compute it. [2]

Recall the definition of a stationary distribution: (Def 7.2.4 of lecture notes)

Definition 7.2.4. Consider a Markov chain $X = \{X_t\}_{t \in \mathbb{N}}$ defined on a discrete state space S with transition matrix $P = [p_{ij}]$. The distribution represented by the vector $\underline{\pi}$ is a stationary distribution of X if $\underline{\pi}$ satisfies the following:

- $\underline{\pi}$ is a vector of probabilities, i.e. $\pi_i \geq 0$ for all $i \in S$, and $\sum_{i \in S} \pi_i = 1$; and
- $\pi_j = \sum_{i \in S} \pi_i p_{ij}$ for all $j \in S$, i.e. $\underline{\pi} = \underline{\pi} P$.

Suppose a stationary distribution exists, it can be found by solving the system of equations:

$$\underline{\pi} = \underline{\pi} P \quad \sum_{i \in S} \pi_i = 1$$

and if a unique solution exists, then the chain has a uniquely stationary distribution.

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ q & 1-q & 0 \end{pmatrix}, \quad q \in (0, 1]$$

Let $\underline{\pi} = (\pi_1, \pi_2, \pi_3)$ be a stationary distribution, then:

$$\underline{\pi} = \underline{\pi} P, \quad \pi_1 + \pi_2 + \pi_3 = 1, \quad \pi_i \geq 0$$

$$\begin{pmatrix} \pi_1 \\ \pi_2 \\ \pi_3 \end{pmatrix} = (\pi_1, \pi_2, \pi_3) \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ q & 1-q & 0 \end{pmatrix} = \begin{pmatrix} q\pi_3 & \textcircled{1} \\ \pi_1 + (1-q)\pi_3 & \textcircled{2} \\ \pi_2 & \textcircled{3} \end{pmatrix}$$

Solve for q :

$$\pi_1 = q\pi_3 \quad \pi_3 = \pi_2$$

$$\pi_1 + \pi_2 + \pi_3 = q\pi_3 + \pi_2 + \pi_2 = q\pi_2 + \pi_2 + \pi_2 = (q+2)\pi_2 = 1$$

$$\pi_2 = \frac{1}{q+2}$$

$$\pi_1 = q\pi_3 = q\pi_2 = q \cdot \frac{1}{q+2} = \frac{q}{q+2}$$

$$\pi_3 = \pi_2 = \frac{1}{q+2}$$

$$\therefore \underline{\pi}(q) = \left(\frac{q}{q+2}, \frac{1}{q+2}, \frac{1}{q+2} \right), \quad q \in (0, 1]$$

- c) (c) For what values of $q \in (0, 1]$ will the sequence of marginal distributions $\pi^{(t)}$ of $\{X_t\}_{t \in \mathbb{N}}$ converge to a unique stationary distribution for all initial distributions $\pi^{(0)}$? [3]

Theorem 7.2.6 (Theorems 6.4.17, 6.4.21, Grimmet and Stirzaker (2001)).

Let $X = \{X_t\}_{t \in \mathbb{N}}$ be a Markov chain defined on a discrete state space S , with n -step transition probabilities denoted $p_{ij}^{(n)}$, $i, j \in S$. For $i, j \in S$, if j is aperiodic, then the n -step transition probability $p_{ij}^{(n)}$ converges as $n \rightarrow \infty$

$$p_{ij}^{(n)} \rightarrow f_{ij}/\mu_j \quad \text{as } n \rightarrow \infty.$$

If we also assume that S is irreducible, i.e. that the Markov chain X is irreducible and aperiodic, then

$$p_{ij}^{(n)} \rightarrow 1/\mu_j \quad \text{as } n \rightarrow \infty \quad \forall i, j \in S.$$

If, furthermore, we assume that the chain is also non-null persistent, i.e. that it is ergodic, then

$$p_{ij}^{(n)} \rightarrow \pi_j = 1/\mu_j \quad \text{as } n \rightarrow \infty \quad \forall i, j \in S,$$

Recall that theorem 7.2.6 from the lecture notes tells us that if a chain is irreducible, aperiodic and ergodic, then:

$$p_{ij}^{(n)} \rightarrow \pi_j = 1/\mu_j \quad \text{as } n \rightarrow \infty \quad \forall i, j \in S.$$

If $p_{ij}^{(n)} \rightarrow \pi_j \quad \forall i, j$, then:

$$\pi_j^{(0)} = \sum_i \pi_i^{(0)} p_{ij}^{(0)} \xrightarrow{0 \rightarrow \infty} \sum_i \pi_i^{(0)} \pi_j = \pi_j \leq \pi_i^{(0)} = \pi_j$$

$\therefore \pi^{(0)} \rightarrow \pi$ for all initial distributions.

So we need to find the values of $q \in (0, 1]$ where $\{X_t\}_{t \in \mathbb{N}}$ is irreducible, aperiodic and ergodic.

From (a), for all $q \in (0, 1]$, $\{X_t\}_{t \in \mathbb{N}}$ is irreducible.

From lecture notes (Remark P120), all states in a finite, irreducible state space are positive recurrent, i.e. non-null persistent.

So we only need to check if aperiodic.

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ q & 1-q & 0 \end{pmatrix}$$

Recall definition 7.2.13 from the lecture notes:

Definition 7.2.13. Consider a Markov chain with transition matrix $P = [p_{ij}]$, defined on the discrete state space S . The *period* of a state $i \in S$, denoted d_i , is defined to be the greatest common divisor of all the possible numbers of iterations in which return to i is possible, i.e.

$$d_i = \gcd\{n : p_{ii}^{(n)} > 0\}.$$

If $d_i > 1$, then state i is said to be *periodic*; if $d_i = 1$, then state i is *aperiodic*.

In an irreducible chain, all states share the same period, so we only need to look at 1 case.

State 1.

we can return in 3 or 5 steps,

$1 \rightarrow 2 \rightarrow 3 \rightarrow 1$,

$$p_{11}^{(3)} = p_{12} p_{23} p_{31} = 1 \cdot 1 \cdot q = q$$

or $1 \rightarrow 2 \rightarrow 3 \rightarrow 2 \rightarrow 3 \rightarrow 1$,

$$p_{11}^{(5)} = p_{12} p_{23} p_{32} p_{23} p_{31} = 1 \cdot 1 \cdot (1-q) \cdot 1 \cdot q = q(1-q)$$

check $0 < q < 1$:

$q > 0$ and $1-q > 0$,

$$\therefore p_{11}^{(3)} > 0 \text{ and } p_{11}^{(5)} > 0$$

$$\therefore d_1 = \gcd(3, 5) = 1$$

$\therefore$ state 1 is aperiodic, and the entire chain is aperiodic by irreducibility (all irreducible chains have the same period)

$\therefore$ by theorem 7.2.6, $p_{ij}^{(n)} \rightarrow \pi_j$ and therefore $\pi^{(n)} \rightarrow \pi$ for initial distributions for $q \in (0, 1)$

Now check $q = 1$

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

check state 1:

$1 \rightarrow 2 \rightarrow 3 \rightarrow 1 \rightarrow 2 \rightarrow 3 \dots$

state 1 (and therefore all states) return at $n = 3, 6, 9, \dots$ (multiples of 3)

so

$$d_1 = \gcd\{n : p_{11}^{(n)} > 0\} = \gcd\{3, 6, 9, \dots\} = 3$$

so chain is periodic with period 3, i.e. marginals cycle and do not converge when $q = 1$.

$\therefore \pi^{(k)}$ converges to a unique stationary distribution for all initial distribution when $0 < q < 1$ or for all $q \in (0, 1)$

Q3) There are up to two marks available in this question for clarity of exposition.

Let $X_1, \dots, X_n$ be independent and identically distributed random variables from the $\text{Uniform}(0, \theta)$ distribution, where $\theta > 0$ is an unknown parameter.

(a) Prove that $M := \max_{i \in \{1, \dots, n\}} X_i$ is a sufficient statistic for θ . Is $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ a sufficient statistic for θ ? Briefly justify your answer. [3]

$$X_i \sim \text{Uniform}(0, \theta), \theta > 0$$

so density of X_i is:

$$f(x|\theta) = \begin{cases} \frac{1}{\theta} & 0 \leq x \leq \theta \\ 0 & \text{otherwise} \end{cases}$$

joint density is:

$$f(x_1, \dots, x_n|\theta) = \begin{cases} \frac{1}{\theta^n} & 0 \leq \max\{x_1, \dots, x_n\} \leq \theta \\ 0 & \text{otherwise} \end{cases}$$

We can rewrite the joint density using the factorisation theorem:

$$\begin{aligned} f(x_1, \dots, x_n|\theta) &= \prod_{i=1}^n \frac{1}{\theta} \mathbf{1}_{\{0 \leq x_i \leq \theta\}} \\ &= \frac{1}{\theta^n} \mathbf{1}_{\{\max x_i \leq \theta\}} \end{aligned}$$

and factorise as $h(x) g(T(x), \theta)$ where $h(x) = 1$ $g(T(x), \theta) = \frac{1}{\theta^n} \mathbf{1}_{\{\max x_i \leq \theta\}}$ and $T(x) = \max x_i$

$\therefore$ by factorisation theorem, T is a sufficient statistic for θ

$s(x) = \max\{x_1, \dots, x_n\}$ is the minimally sufficient statistic for $\text{Uniform}(0, \theta)$.

let $T(x) = \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$, since there does not exist a function such that $f(T(x)) = s(x)$, then $T(x) = \bar{X}$ cannot be a sufficient statistic.

- (b) Write down the likelihood $L(\theta | x)$ of θ and use it to derive the maximum likelihood estimator $\hat{\theta}_{MLE}$ of θ . [3]

$$L(\theta | x) = f_x(x | \theta) = \frac{1}{\theta^n} \mathbb{1}_{\{\max\{x_1, \dots, x_n\} \leq \theta\}}$$

$\hat{\theta}_{MLE}$ maximises the likelihood

if $\theta < M$, then at least one x_i lies outside $(0, \theta)$, so the density is 0.

if $\theta \geq M$:

$$L(\theta | x) = \theta^{-n}$$

This is strictly decreasing, so the value that maximises the likelihood is the smallest value of θ , but $\theta \geq M$, so $\theta = M$.

$$\therefore \hat{\theta}_{MLE} = M = \max\{x_1, \dots, x_n\}$$

- (c) Derive the method-of-moments estimator $\hat{\theta}_{MoM}$ of θ . [2]

Set the moments of the distribution equal to the sample moments.

$$\mu := E[X] = \frac{\theta}{2}$$

$$\bar{X} = \frac{\theta}{2} \Rightarrow \hat{\theta}_{MoM} = 2\bar{X}$$

- (d) Derive the CDF of $\hat{\theta}_{MLE}$ and use it to construct an interval estimator for θ from $\hat{\theta}_{MLE}$ with exactly 95% coverage probability. [4]

$$\hat{\theta}_{MLE} = M = \max\{x_1, \dots, x_n\}$$

$$F_{\hat{\theta}}(t) = P_{\theta}(M \leq t) = P_{\theta}(x_1 \leq t, \dots, x_n \leq t) \quad \text{for any } t$$

$$P_{\theta}(x_i \leq t) = \begin{cases} 0, & t \leq 0 \\ \frac{t}{\theta}, & 0 < t < \theta \\ 1, & t \geq \theta \end{cases}$$

CDF of $\hat{\theta}_{MLE}$:

$$\therefore F_{\theta}(t) = \begin{cases} 0, & t \leq 0 \\ \left(\frac{t}{\theta}\right)^n, & 0 < t < \theta \\ 1, & t \geq \theta \end{cases}$$

Now construct interval estimator:

$$P_{\theta}(\theta \in [M, yM]) = P_{\theta}(\theta \leq yM) = P_{\theta}\left(M \geq \frac{\theta}{y}\right) = 1 - F_{\theta}\left(\frac{\theta}{y}\right) = 1 - \left(\frac{1}{y}\right)^n$$

$$1 - \left(\frac{1}{y}\right)^n = 0.95$$

$$\rightarrow \left(\frac{1}{y}\right)^n = 0.05 = \frac{1}{20}$$

$$y^{-n} = \frac{1}{20}$$

$$y = 20^{\frac{1}{n}} = \sqrt[n]{20}$$

so the 95% confidence interval based on the MLE is

$$[\hat{\theta}_{MLE}, 20^{\frac{1}{n}} \hat{\theta}_{MLE}] = [M, 20^{\frac{1}{n}} M]$$

(e) Use the central limit theorem to derive an interval estimator for θ from $\hat{\theta}_{MoM}$ with approximately 95% coverage probability. [4]

You may use without proof that for a random variable $X \sim \text{Uniform}(0, \theta)$,

$$\mu := \mathbb{E}[X] = \frac{\theta}{2}, \quad \sigma^2 := \text{Var}[X] = \frac{\theta^2}{12}.$$

For a random variable $Z \sim N(0, 1)$, the table below shows the values of the quantiles z_{α} such that

$$P(Z \leq z_{\alpha}) = \alpha,$$

for selected values of α .

α	0.9	0.95	0.975	0.99	0.995	0.999
z_{α}	1.282	1.645	1.960	2.326	2.576	3.090

[18]

Recall that $\hat{\theta}_{mom} = 2\bar{X}$

For $X_i \sim \text{Uniform}(0, \theta)$

$$\mathbb{E}[X_i] = \frac{\theta}{2}, \quad \text{Var}(X_i) = \frac{\theta^2}{12}$$

$$\mathbb{E}[\bar{X}] = \frac{\theta}{2}, \quad \text{Var}(\bar{X}) = \text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{n}{n^2} \frac{\theta^2}{12} = \frac{1}{n} \frac{\theta^2}{12} = \frac{\theta^2}{12n}$$

$$E[\hat{\theta}_{\text{mom}}] = 2 \cdot E[\bar{x}] = \theta$$

$$\text{Var}[\hat{\theta}_{\text{mom}}] = 2^2 \cdot \text{Var}(\bar{x}) = 4 \cdot \frac{\theta^2}{12n} = \frac{\theta^2}{3n}$$

$$\frac{\hat{\theta}_{\text{mom}} - E[\hat{\theta}_{\text{mom}}]}{\sqrt{\text{Var}(\hat{\theta}_{\text{mom}})}} = \frac{\hat{\theta}_{\text{mom}} - \theta}{\sqrt{\theta^2/3n}} = \frac{\hat{\theta}_{\text{mom}} - \theta}{\theta/\sqrt{3n}} \approx N(0,1) \text{ as } n \rightarrow \infty \text{ (by CLT)}$$

Using 0.975 quantile from the table, $z_c = 1.96$ (two sided CI so $P(-z_{0.975} \leq z \leq z_{0.975}) = 0.95$)

$$P\left(-1.96 \leq \frac{\hat{\theta}_{\text{mom}} - \theta}{\theta/\sqrt{3n}} \leq 1.96\right) \approx 0.95$$

$$P\left(\frac{-1.96}{\sqrt{3n}} \leq \frac{\hat{\theta}_{\text{mom}} - \theta}{\theta} \leq \frac{1.96}{\sqrt{3n}}\right)$$

$$= P\left(\frac{-1.96}{\sqrt{3n}} \leq \frac{\hat{\theta}_{\text{mom}}}{\theta} - 1 \leq \frac{1.96}{\sqrt{3n}}\right)$$

$$= P\left(1 - \frac{1.96}{\sqrt{3n}} \leq \frac{\hat{\theta}_{\text{mom}}}{\theta} \leq 1 + \frac{1.96}{\sqrt{3n}}\right)$$

$$= P\left(\frac{\theta}{1 + \frac{1.96}{\sqrt{3n}}} \geq \hat{\theta}_{\text{mom}} \geq \frac{\theta}{1 - \frac{1.96}{\sqrt{3n}}}\right) \approx 0.95$$

So the approximate CI for θ from $\hat{\theta}_{\text{mom}}$ is :

$$CI = \left[\frac{\theta}{1 + \frac{1.96}{\sqrt{3n}}}, \frac{\theta}{1 - \frac{1.96}{\sqrt{3n}}} \right]$$

Q4) Three widely used measures of distance between probability distributions are the *Total Variation*, the *Kullback-Leibler divergence* and the *Hellinger distance*. Consider two probability measures P, Q defined on a common measurable space $(\Omega, \mathcal{F})$. The *Total-Variation* is defined as

$$\text{TV}(P, Q) := \sup_{A \in \mathcal{F}} |P(A) - Q(A)|.$$

If Ω is discrete and $P(x), Q(x) > 0$ for all $x \in \Omega$, then the *Kullback-Leibler divergence* is defined as

$$\text{KL}(P, Q) = \sum_{x \in \Omega} P(x) \log \frac{P(x)}{Q(x)}.$$

If P, Q are absolutely continuous with respect to some reference measure, so that they have probability density function p, q , respectively, then the *squared Hellinger distance* is defined as

$$H^2(P, Q) = \frac{1}{2} \int_{\Omega} (\sqrt{p(x)} - \sqrt{q(x)})^2 dx.$$

Answer the following questions assuming the settings in which each of the measures of distance are defined above.

a) Is Total Variation a metric? Justify your answer.

[3]

Recall Definition 10.1.1 in the lecture notes:

Definition 10.1.1. Let S be a set, and consider the bivariate *distance function* $d : S \times S \rightarrow [0, \infty)$. We define d to be a *metric* on S if it satisfies the following three conditions:

i) $d(x, y) = 0 \iff x = y$

[Identity of indiscernibles]

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10. Distance Functions

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ii) $d(x, y) = d(y, x)$

[Symmetry]

iii) $d(x, z) \leq d(x, y) + d(y, z)$

[Triangle inequality]

and we further define the pair (S, d) to be a *metric space*.

Let P, Q be probability measures defined on $(\Omega, \mathcal{F})$

$$\text{TV}(P, Q) := \sup_{A \in \mathcal{F}} |P(A) - Q(A)|$$

Test for axioms:

Identity of indiscernibles:

if $TV(P, Q) = 0$,

$$|P(A) - Q(A)| \leq \sup_B |P(B) - Q(B)| = 0 \quad \text{for all } A$$

$$\therefore P(A) = Q(A) \quad \forall A \Rightarrow P = Q$$

if $P = Q$:

$$P(A) - P(A) = 0 \quad \forall A \Rightarrow TV(P, Q) = 0$$

$$\therefore TV(P, Q) = 0 \Leftrightarrow P = Q$$

Symmetry:

$$|P(A) - Q(A)| \geq 0 \quad \forall A \Rightarrow \sup_A |P(A) - Q(A)| \geq 0 \Rightarrow TV(P, Q) \geq 0$$

$$|P(A) - Q(A)| = |Q(A) - P(A)| \Rightarrow \sup_A |P(A) - Q(A)| = \sup_A |Q(A) - P(A)|$$

$$\therefore TV(P, Q) = TV(Q, P)$$

Triangle inequality

Let R be a probability measure on $(\Omega, \mathcal{F})$

For any $A \in \mathcal{F}$,

$$|P(A) - R(A)| \leq |P(A) - Q(A)| + |P(A) - R(A)|$$

$$TV(P, R) = \sup_A |P(A) - R(A)| \leq \sup_A |P(A) - Q(A)| + \sup_A |P(A) - R(A)| = TV(P, Q) + TV(Q, R)$$

$\therefore$ As TV satisfies all three axioms it is a metric

b) Is Kullback-Leibler a metric? Justify your answer.

No, by counterexample. Take $\Omega = \{1, 2\}$, $P = \{\frac{1}{2}, \frac{1}{2}\}$, $Q = \{\frac{1}{4}, \frac{3}{4}\}$

$$KL(P, Q) = \frac{1}{2} \log \frac{\frac{1}{2}}{\frac{1}{4}} + \frac{1}{2} \log \frac{\frac{1}{2}}{\frac{3}{4}} = \frac{1}{2} \log 2 + \frac{1}{2} \log \frac{2}{3} = \frac{1}{2} \log \frac{4}{3}$$

$$KL(Q, P) = \frac{1}{4} \log \frac{\frac{1}{4}}{\frac{1}{2}} + \frac{3}{4} \log \frac{\frac{3}{4}}{\frac{1}{2}} = \frac{1}{4} \log \frac{1}{2} + \frac{3}{4} \log \frac{3}{2} = \frac{1}{4} \log \frac{3}{4} \neq \frac{1}{2} \log \frac{4}{3}$$

$\therefore KL(Q, P) \neq KL(P, Q)$ as it does not pass symmetry by counter-example

c) By giving a counter example, show that the squared Hellinger distance is *not* a metric.
[3]

$$H^2(P, Q) = \frac{1}{2} \int_{\Omega} (\sqrt{p(x)} - \sqrt{q(x)})^2 dx$$

Take $\Omega = \{1, 2\}$ a discrete space with counting measure $\mu(1) = \mu(2) = 1$, define distributions $P = (1, 0)$, $Q = (\frac{1}{2}, \frac{1}{2})$, $R = (0, 1)$

$$\text{then } \int_{\Omega} f(x) d\mu = \sum_{\Omega} f(x)$$

$$\begin{aligned} H^2(P, Q) &= \frac{1}{2} \sum_{i=1}^2 (\sqrt{p(x)} - \sqrt{q(x)})^2 \\ &= \frac{1}{2} \left[(1 - \sqrt{\frac{1}{2}})^2 + (0 - \sqrt{\frac{1}{2}})^2 \right] \\ &= \frac{1}{2} \left[1 - \sqrt{2} + \frac{1}{2} + \frac{1}{2} \right] = 1 - \frac{\sqrt{2}}{2} \end{aligned}$$

$$\begin{aligned} H^2(Q, R) &= \frac{1}{2} \sum_{i=1}^2 (\sqrt{q(x)} - \sqrt{r(x)})^2 \\ &= \frac{1}{2} \left[\left(\sqrt{\frac{1}{2}} - 0 \right)^2 + \left(\sqrt{\frac{1}{2}} - 1 \right)^2 \right] \\ &= \frac{1}{2} \left[\frac{1}{2} + \frac{1}{2} - \sqrt{2} + 1 \right] = 1 - \frac{\sqrt{2}}{2} \end{aligned}$$

$$\begin{aligned} H^2(P, R) &= \frac{1}{2} \sum_{i=1}^2 (\sqrt{p(x)} - \sqrt{r(x)})^2 \\ &= \frac{1}{2} \left[(1 - 0)^2 + (0 - 1)^2 \right] = 1 \end{aligned}$$

triangle inequality:

$$H^2(P, R) \leq H^2(P, Q) + H^2(Q, R)$$

$$1 > 1 - \frac{\sqrt{2}}{2} + 1 - \frac{\sqrt{2}}{2} = 2 - \sqrt{2}$$

$\therefore$ the triangle inequality fails, and so the squared Hellinger distance is not a metric, by counterexample.